Fiscal Year:	FY 2012	**Task Last Updated:**	FY 11/30/2012
PI Name:	Cornforth, Michael Ph.D.		
Project Title:	Structural Chromosome Aberrations Formed in Response to Changes in Proton Energy and Dose Rate		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77555-5302	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: NCE until 8/31/2012, per C. Guidry/JSC (Ed., 9/26/2011) NOTE: Received NCE until 8/31/2011 (original end date was 8/31/2010), per C. Guidry/JSC (10/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Loucas, Bradford (University of Texas Medical Branch)		
Grant/Contract No.:	NNX07AP83G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	From the perspective of NASA, risk uncertainties related to radiation exposure during space exploration are unacceptably large. These include the effects of sporadic solar particle events (SPEs) which can produce relatively high levels of radiation, principally from protons. The resultant dose to personnel from an SPE can be high enough to warrant concerns of cancer induction. Because human epidemiological data in this area are not available, reduction in risk uncertainties will require data from meaningful surrogate endpoints, such as chromosome damage. A reasonable and widely held notion is that dose rate effects from protons will be the same as for x- and gamma rays. There is, however, not much direct experimental evidence to support this contention, which adds additional uncertainty to risk projections. We propose to simulate an SPE using proton emissions from the National Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL). Owing to differences in ionization density that may translate to differences in biological effect, proton energies of 50 MeV and 250 MeV will be studied. These will be delivered at both high and low dose rates. Results will then be compared to those obtained for gamma rays from a 137-Cesium source. Much is known about chromosomal response to other types of radiation, such as x- and gamma rays, including how the level of damage is influenced by the rate at which radiation is delivered. We have chosen structural damage to chromosomes of human cells in culture as a surrogate endpoint for cell death and cancer. We first propose to construct quantitative dose responses for the induction of chromosome aberrations found in cells when they arrive at the first mitosis following exposure. For this and other objectives, multiplex fluorescence in situ hybridization (mFISH) will be used, which effectively paints each of the 23 pairs of human chromosomes a unique color. This allows even very complicated rearrangements among all chromosomes to be visualized simultaneously. The second set of objectives involves analysis of chromosome damage in clonal descendents of single cells that survive exposure to radiation. Several such clones will be expanded in culture and examined by mFISH for transmissible chromosome aberrations. A subset of these clones will be further examined for structural aberrations that are too small to be seen under the microscope, by employing comparative genomic hybridization applied to DNA microarrays (aCGH). It is argued that genetic damage in cells surviving radiation may be diagnostic of the type of radiation encountered, and that such damage has more direct relevance to carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overarching objective of these studies is to help reduce uncertainties associated with exposure to ionizing radiation in a space environment, especially those associated with exposure to solar particle events (SPEs) that relate chiefly to high energy proton exposure. Currently the assumption is that such protons behave essentially like other forms of low LET (linear energy transfer) photons such as X- and gamma rays, for which much quantitative information already exists. The extent to which the clastogenic (chromosome aberration producing) effects of protons actually are (or are not) similar to photons in the context of energy and dose rate is an important piece of information needed to help reduce the aforementioned uncertainties.
Task Progress:	Taskbook Report #NNX07AP83G- M.N. Cornforth Our objectives for this project relate to quantifying the clastogenic effects of changes in instantaneous dose rates for energetic protons. On the basis of biophysical modeling, and constraints of NSRL facilities, we determined energies and dose rates that were compatible with our scientific objectives. We made two runs at NSRL. For the spring run we used 150 MeV protons and 1 GeV protons, both of which were delivered at high (30 cGy/min) and low (30 cGy/h) dose rates to telomerase-immortalized human fibroblasts to total absorbed doses of (not including the sham-irradiated zero-dose control) 100, 200, and 300 cGy. In the fall run 50 MeV protons became available, and this energy was used instead of the previous 150 MeV treatment group. As in the spring, 1 GeV protons were used again, which served as a repeat. For both NSRL runs, density-inhibited telomerase-positive BJ-1 human fibroblasts were re-fed with conditioned medium prior to shipment to NRSL. The morning following exposure, flasks were shipped back to UTMB for immediate subculture. Overall final progress for this project can be summarized as follows. The expected diminution of chromosome damage per unit dose following reduction in dose rate was confirmed, as was the predicted attenuation in upward curvature. These results are consistent with the assumption that protons and photons elicit a qualitatively similar biological response. Curvature in the dose response following 150 MeV protons was effectively eliminated at the LDR of 0.05 cGy/min. This is evidence that the limiting low dose rate (LLDR) was reached, and is consistent with expectations based on gamma ray studies. Irrespective of energy, the preponderance of curvature in all responses derives from complex exchanges, which is also similar to the situation for photons. The above conclusions regarding similarity of effect between high energy protons and gamma rays need to be tempered by the following observations: The unexpected observation that higher energy protons were more effective at producing chromosome damage per unit dose. [More data for the 50 MeV protons (see completion matrix) will be needed to establish whether this is a trend.] The fact that curvature in the response was not altogether eliminated for the higher energy 1 GeV protons following LDR. [It is unknown whether the LLDR hadn’t yet been reached for this radiation, or whether single track action is capable of producing such curvature.] Additional work may eventually be needed to put hard numbers on quantitative differences regarding gamma photons versus protons in order to estimate meaningful RBEs. [In particular this would include more dose points over the lower dose range.] Regarding clones recovered from exposed cell populations, roughly 25% following 3 Gy of protons harbored transmissible chromosome rearrangements, most of which were simple reciprocal translocations; a significant number contained complex exchanges as well that involve simultaneously three or more chromosomes. These results are consistent with those observed following exposure to 4 Gy 137Cs gamma-rays. Because they are homogeneous for the particular rearrangements they carry, these clonal derivatives serve as ideal material for molecular analysis, as for example, analysis by array-based comparative genomic hybridization (aCGH) below. We used high-density CGH arrays (aCGH) to examine the distribution of large (megabase-sized) and small (tens of kilobase size) deletions in several of the surviving clones described above. The size-frequency distribution of such deletions, on both the megabase and kilobase scales suggest that the formation of smaller deletions are favored over larger ones, but that the distributions are characterized as having “long asymmetrical tails.” In at least one case a clone had suffered a deletion in near proximity to an apparently-simple reciprocal translocation, suggesting the rearrangements

	was “complex,” i.e., a codeletion, as part of the translocation/exchange process. In collaboration with the Oregon Health and Science University we have begun to explore the usefulness of pair-end whole-genome sequencing to analyze structural rearrangements in various clones that survived exposure to either gamma rays or 56-Fe ions. Additional funding would be required to determine the feasibility of this approach.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
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Articles in Peer-reviewed Journals	Loucas BD, Cornforth MN. "The LET dependence of unrepaired chromosome damage in human cells: A break too far?" Radiat Res. 2013 Apr;179(4):393-405. (Ed. note: originally reported as 'Accepted as of December 2012.') http://dx.doi.org/10.1667/RR3159.2 ; PubMed PMID: 23578187 , Apr-2013